

Problem Set 20

Problem 16.2.4: The Coulomb repulsion between a $Z = 90$ nucleus and an α particle (charge $2e$ and mass m) is described by the potential

$$V(x) = \begin{cases} \frac{2Ze^2}{x} & \text{if } x \geq x_0, \\ 0 & \text{if } 0 \leq x \leq x_0, \end{cases}$$

where the last line describes (very badly) the interior of the nucleus, and $x_0 = 10^{-12}$ cm. For an energy $E = 4.2$ MeV, the classical turning points are at $x = x_0$ and $x = x_e$, where $E = 2Ze^2/x_e$, implying $x_e = 2Ze^2/E$. Note that we can write $V(x) = x_e E/x$ for $x \geq x_0$. Then

$$\begin{aligned} \gamma &= \frac{2}{\hbar} \int_{x_0}^{x_e} \sqrt{2m[V(x) - E]} dx = \frac{2\sqrt{2mE}}{\hbar} \int_{x_0}^{x_e} \sqrt{\frac{x_e}{x} - 1} dx = \frac{2x_e\sqrt{2mE}}{\hbar} \int_a^1 \sqrt{\frac{1}{y} - 1} dy \\ &= \frac{2Ze^2}{\hbar v} \left[\pi + 2 \tan^{-1} \left(\frac{\sqrt{1-a}(1-2a)}{2\sqrt{a}(1-a)} \right) - 4\sqrt{a(a-1)} \right] \end{aligned}$$

where

$$\begin{aligned} \frac{v}{c} &\equiv \sqrt{\frac{2E}{mc^2}} = \sqrt{\frac{2(4.2 \times 10^6 \text{eV})}{4(9.4 \times 10^8 \text{eV})}} = 0.047, \\ a &\equiv \frac{x_0}{x_e} = \frac{x_0 E}{2Ze^2} = \frac{x_0 E}{2Z} \frac{1}{\hbar c} \frac{\hbar c}{e^2} = \frac{(10^{-12} \text{cm})(4.2 \times 10^6 \text{eV})(140)}{2(90)(2 \times 10^3 \text{eV})(10^{-8} \text{cm})} = 0.16. \end{aligned}$$

Plugging these into γ gives $\gamma = 88$. Therefore the half-life, τ , is

$$\tau = \frac{1}{R} = \frac{2x_0}{(v/c)c} e^\gamma = \frac{(2 \times 10^{-12} \text{cm})e^{88}}{(0.047)(3 \times 10^{10} \text{cm/s})} = 2 \times 10^{17} \text{s} \approx 10^{10} \text{years}.$$

Problem 16.2.5: With the linear potential $V(r) = V_0 + kr$ for $r > 0$ and with a vertical wall at $r = 0$ ($V(0) = \infty$), the WKB approximation gives for the n th level

$$(n + \frac{3}{4})\hbar\pi = \int_0^{r_n} \sqrt{2\mu(E_n - V_0 - kr)} dr, \quad (1)$$

where $\mu = (m + m)/2m = m/2$ is the reduced mass, the shift of $\frac{3}{4}$ is because $\ell = 0$ and there is the vertical wall at $r = 0$, and the classical turning point r_n is given by $E_n = V(r_n)$ which implies

$$r_n = (E_n - V_0)/k.$$

So (1) implies $(n + \frac{3}{4})\hbar\pi = \sqrt{mk} \int_0^{r_n} \sqrt{r_n - r} dr = \frac{2}{3}\sqrt{mk}r_n^{3/2}$ which implies (squaring)
 $(n + \frac{3}{4})^2\hbar^2\pi^2 = \frac{4}{9}(m/k^2)(E_n - V_0)^3$, giving

$$E_n = V_0 + \left(n + \frac{3}{4}\right)^{2/3} \alpha, \quad \text{where} \quad \alpha \equiv \left(\frac{9k^2\hbar^2\pi^2}{4m}\right)^{1/3}. \quad (2)$$

In the problem we are given the masses (rest energies) $E_0 = 3.1$ GeV (corresponding to the $n = 0$ level) and $E_1 = 3.7$ GeV (corresponding to the $n = 1$ level). Plugging into (2) and solving for V_0 and α , we find $V_0 = 2.3$ GeV, and $\alpha = 0.96$ GeV. As a check, use these values with $n = 2$ and predict $E_2 = 4.2$ GeV, in agreement with the value given in the problem.